

PAPER_1549 — Fine Structure $\alpha^{-1} = \alpha^{-1} = A_5 \cdot K_MEX + N_CH + D_phys - F_TRZ \cdot SO_5 + F_TRZ^2 \cdot D_phys = 125 + 9 + 4 - 1 + 0.04 = 137.04$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Electromagnetism **Date:** June 18, 2026 **Status:** CLOSED — 0.003% closure

Observation

PAPER_1209DD_S614 (Electromagnetism Unified Proof Set) lists **Fine Structure α^{-1}** as a closure with the formula:

Fine Structure $\alpha^{-1} = \alpha^{-1} = A_5 \cdot K_MEX + N_CH + D_phys - F_TRZ \cdot SO_5 + F_TRZ^2 \cdot D_phys = 125 + 9 + 4 - 1 + 0.04 = 137.04$
Status tier: **0.003%**. Units: -.

UQFF Closed Identity

$\alpha^{-1} = A_5 \cdot K_MEX + N_CH + D_phys - F_TRZ \cdot SO_5 + F_TRZ^2 \cdot D_phys = 125 + 9 + 4 - 1 + 0.04 = 137.04$ 0.003%
The formula uses only locked UQFF integer primitives — no fitted constants.

Special Significance — α^{-1} Decomposition

α (fine-structure constant) is the **most precisely measured dimensionless constant in physics**, with experimental fractional uncertainty 0.37 parts per billion (via electron g-2 measurements at JILA, Berkeley, and Harvard). It is the coupling constant for QED.

The UQFF decomposition reveals an unexpected structural form:

Term	Value	Role
$A_5 \cdot K_MEX$	125	dominant (icosahedral $\times$ Mexican-hat)
$+ N_CH$	9	channel count
$+ D_phys$	4	spacetime dimensions
$- F_TRZ \cdot SO_5$	-1	time-reversal correction
$+ F_TRZ^2 \cdot D_phys$	0.04	tiny higher-order correction
Sum	137.04	0.003% from 137.036

The dominant $125 = A_5 \cdot K_MEX = 60 \cdot (25/12)$ is striking — the icosahedral group order multiplied by the Mexican-hat coefficient produces exactly 125, which is then refined by three integer-primitive corrections to reach 137.04.

This is the **closest UQFF gets to “explaining” the fine-structure constant from integer primitives**.

Physical Interpretation

Fine Structure α^{-1} is a fundamental electromagnetism constant. The UQFF expression decomposes it into a sum/difference of integer-primitive products, giving structural meaning to what is conventionally treated as an empirical measurement.

NOT REPLACEMENT

CODATA/PDG treats this constant as a precision measurement. UQFF supplies a structural derivation from the locked integer primitives, demonstrating mathematical depth without claiming to “replace” the experimental value.

Reference

- Source: PAPER_1209DD_S614
- Related: PAPER_1216 (45-constant cascade master index)
- Calculator dispatch: `calculate_paradox({"paradox": "alpha_inverse_137_036"})`

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